

Between driver and driverless: modelling remote operators for autonomous ride-pooling in MATSim

Paper submitted for the MATSim user meeting 2026

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Keywords: ride-pooling; autonomous mobility-on-demand; remote operation; teleoperation; driver shifts; DRT; MATSim.

1. Introduction

Autonomous mobility-on-demand (AMoD) is widely expected to lower the operating cost of ride-pooling services by removing the on-board driver. Yet current robotaxi deployments show that the driver is not eliminated but *displaced*: fleets still rely on remote operators who supervise vehicles and intervene whenever the automated driving system (ADS) encounters a situation beyond its operational design domain. One operator now supervises many vehicles (1: N) rather than being bound one-to-one to a single vehicle, but the labour does not disappear, it is restructured. Reported supervision ratios today range from roughly 1:3 for early-stage operators to about 1:43 for the market leader, and the associated wage bill scales linearly with fleet size at a fixed ratio.

Simulation frameworks for on-demand services have so far modelled either the fully manual case, where a driver is bound to each vehicle for an entire shift, or the idealised autonomous case, where vehicles are treated as constraint-free and permanently available. The realistic middle, autonomous vehicles that are still *not* free of operational duties because a finite pool of remote operators must resolve stochastic incidents, has not been represented. This contribution presents a new remote-operator shift module for MATSim that fills exactly this gap. It is a direct continuation of the driver-shift extension of Zwick et al. (2022), reusing its shift, break and hub-return architecture but generalising the notion of a “shift” so that it can also represent an operator supervising a dynamically sized set of vehicles. We describe the module’s design, the way it emits *virtual shifts* to drive the number of concurrently active vehicles, and the stochastic incident process that couples fleet performance to operator availability. We further argue that the resulting operator pool is, structurally, an $M/G/m$ queue, which lets us validate the implementation against closed-form queueing results.

2. Background

MATSim (Horni et al., 2016) simulates demand-responsive transport through its Dynamic Vehicle Routing Problem (DVRP) extension and Demand-Responsive Transport (DRT) module (Bischoff et al., 2017), which together provide request insertion, pooling, routing and dispatching for on-demand fleets. These have been applied to shared autonomous vehicle fleets in numerous urban contexts. Zwick et al. (2022) extended the DRT framework with an explicit model of *operational* shift constraints: vehicles are bound to driver shifts with mandatory breaks and are routed to designated hubs for shift changeovers, breaks and (for electric fleets) charging. Their central finding is that a fleet operating under realistic shift plans serves fewer rides than the same fleet assumed to operate continuously, quantifying the operational efficiency gap between the manual and the idealised autonomous regime.

That work measured the two *ends* of the automation spectrum. Remote guidance is the missing middle. In the autonomous-but-supervised regime the one-driver-per-vehicle coupling is broken: a single operator can supervise κ vehicles at once, so capital (better ADS, fewer interventions) and labour (operators on shift) become partial substitutes rather than strict complements. Modelling this regime requires three ingredients that the driver-shift model does not have: a capacity-gating mechanism that ties the number of active vehicles to operator availability, a stochastic incident process that consumes operator capacity, and a queue that forms when incidents arrive faster than operators can resolve them. Our module adds these three mechanisms while remaining fully interoperable with the existing DRT and shift infrastructure, so that a fleet may even mix conventional driver shifts and remote-operated vehicles in the same simulation.

3. Methodology

The module runs alongside the standard MATSim mobility simulation, the DVRP-based DRT dispatcher and the shift dispatcher of Zwick et al. (2022). It consists of three interacting components (capacity gating, a stochastic incident process and guidance queueing) whose interplay is sketched in Figure 1.

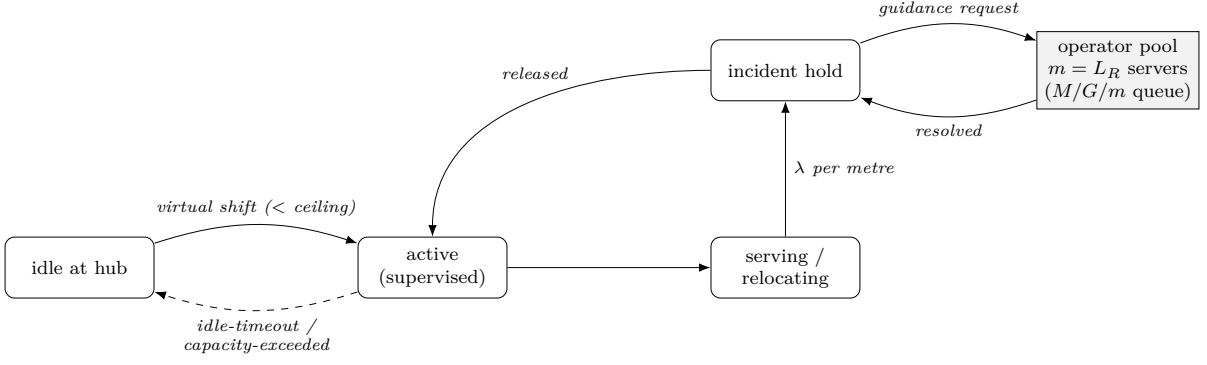


Figure 1: Vehicle and operator life cycle. Idle-at-hub vehicles are activated via transient virtual shifts up to the ceiling $\sum \kappa$; active vehicles generate incidents per driven metre, each occupying one operator from the $M/G/m$ pool, which queues when all are busy.

Capacity gating via virtual shifts. An operator shift is characterised by a start time, an end time and a supervision capacity κ (the maximum number of vehicles one operator may supervise concurrently). The effective fleet capacity, i.e. the number of vehicles permitted to be active at time t , is bounded by operator availability,

$$N_{\text{active}}(t) = \min(N, \kappa \cdot L_R(t)), \quad (1)$$

where N is the total fleet size and $L_R(t)$ the number of operators currently on shift. The key design decision is *how* the module reaches this ceiling. Rather than introducing a parallel activation mechanism, we generalise the shift scheduler of Zwick et al. (2022) to emit **virtual shifts** for idle-at-hub vehicles whenever the active count is below the current ceiling. A virtual shift is processed by the regular shift dispatcher exactly like an ordinary driver shift (it is assigned to an idle vehicle and started), but it carries no committed end: it is transient and lives only within the simulation, and is never written into the persistent shift plan. This reuse means activation, dispatching and hub logistics all flow through the existing, tested shift machinery. A reconciler compares the current supervised count against a target (a demand-following signal clamped to the ceiling $\sum \kappa$) and returns how many new virtual shifts to emit; the same target is read on the deactivation side, so both margins agree, and a trailing-window smoothing of the “busy” signal damps the low-demand activate↔idle-timeout↔recall oscillation. Vehicle activation and deactivation (with a reason such as capacity-exceeded or idle-timeout) are emitted as events for downstream analysis.

Stochastic incident process. Each active vehicle generates incidents according to a Poisson process over the distance it drives, at hazard rate λ per metre, i.e. on average one incident every $1/\lambda$ vehicle-kilometres driven (applied to both occupied and empty kilometres). The base rate is a proxy for ADS maturity: a more capable system needs fewer human interventions per vehicle-kilometre. Incidents are drawn per severity class, each an independent Poisson process, and each class carries its own duration distribution (log-normal by default). When an incident occurs, the vehicle is held and one operator is occupied for the incident’s duration; severity steers only the duration, never how much capacity an incident consumes (an incident always occupies exactly one operator). The incident hold is realised as a regular vehicle task that also participates in the electric-vehicle energy bookkeeping, so incidents work identically for conventional and electric fleets.

Guidance queuing. When incidents arrive faster than operators are free to handle them, guidance requests queue: the affected vehicle stands still and waits until an operator becomes available, and operators clear the queue one request after another. This is the mechanism that couples operator supply to fleet performance: a scarce operator pool lengthens the effective time each vehicle is held, drives up passenger waiting time and, at the extreme, causes trip rejections. All state transitions (incident start and end, activation and deactivation, shift start and end, queue delay) are written to the standard MATSim event stream, from which per-simulation utilisation, incident and operator-hour analyses are produced.

An operator pool as a queue. These mechanisms make the operator pool, structurally, an $M/G/m$ queueing system: $m = L_R$ servers, Poisson arrivals at rate λ times the fleet’s distance rate, and a general (by default log-normal) service-time distribution. This observation is useful for verification. By configuring the incident service time to be exponential, the system reduces to $M/M/m$, for which the Erlang-C formula gives closed-form predictions for the probability of queueing, the mean and full distribution of queue delay, and server utilisation. Reproducing these in a controlled light-load setting (all operators on shift, non-binding κ , a single severity class) verifies the queuing implementation; the deviations that appear once the model’s realistic features are switched back on are examined below.

Where the arrival rate comes from. The most instructive of those deviations concerns the arrival rate itself. Because incidents are generated per driven *metre*, the temporal arrival rate is not exogenous but a product of how far the fleet drives: the per-vehicle hazard per unit time is $\lambda \bar{v}$, so the mean time to the next incident is $1/(\lambda \bar{v})$, and the fleet-wide rate is

$$\Lambda(t) = \lambda \sum_i v_i(t) = \lambda \frac{d}{dt} \text{VKT}(t) \approx \lambda N_{\text{active}}(t) \delta(t) \bar{v}(t), \quad (2)$$

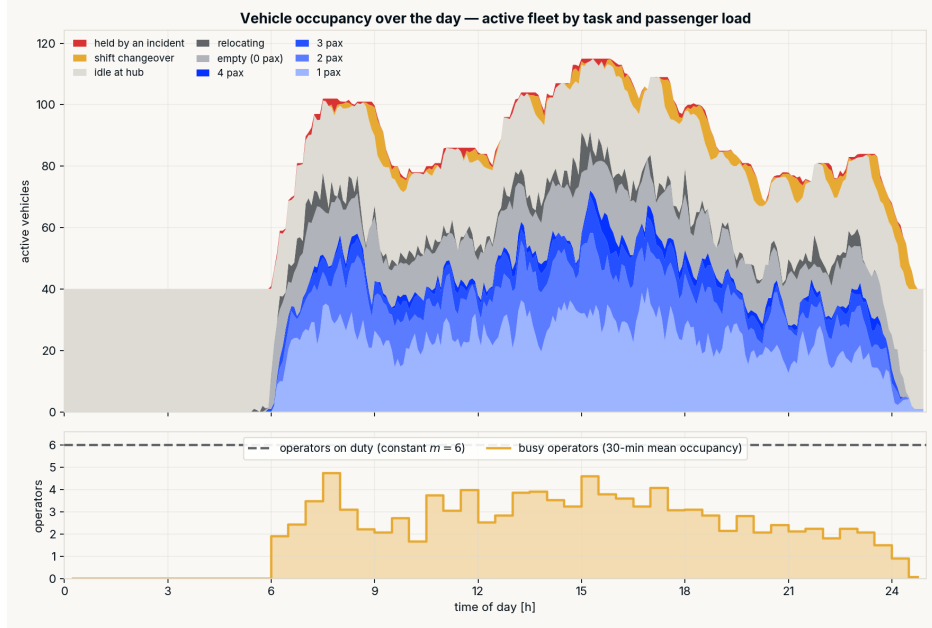


Figure 2: Fleet and operator state over one simulated day (Hamburg; exponential single-class incidents, one per 12 vehicle-km, $m = 6$). *Top:* active-fleet occupancy by passenger load, empty driving, relocation, idle-at-hub, changeover and (red) incident hold. *Bottom:* the $m = 6$ operator pool, its occupancy well below the ceiling.

where $\bar{v}$ is the mean driving speed and δ the share of active time a vehicle actually spends driving (rather than idle, waiting or held). A useful invariant follows: the *number* of incidents over a day equals λ times total VKT and is independent of speed, which changes only *when* incidents occur, not how many. Daily operator load therefore depends on distance and mean duration alone, but *contention* depends on $\bar{v}$, because $\bar{v}$ shapes $\Lambda(t)$. This is where a closed-form queue must borrow from the simulation: writing $\Lambda(t)$ for an Erlang calculation needs $N_{\text{active}}(t)$, $\delta(t)$ and $\bar{v}(t)$, all emergent outputs of demand, routing, pooling, empty mileage, time-varying network speeds and the capacity gate. In the peaks the effect is countervailing (more vehicles active, yet congestion lowers $\bar{v}$), so peak operator demand is not simply peak trip demand: the arrival rate is an emergent, feedback-laden quantity, which is what shapes the deviations we report below.

4. Results and discussion

We exercise the module on an autonomous ride-pooling scenario for Hamburg, Germany, based on empirical MOIA trip demand for a full day and an OpenStreetMap network with time-varying travel times. Vehicles start and end their (virtual) shifts at predefined hub locations and return to hubs when recalled. Figure 2 reports the resulting fleet state over the day. The active fleet is sized to demand through virtual-shift activation: it grows into the morning ramp, peaks around midday and midafternoon, and shrinks again toward the night, always resting on a band of vehicles activated but currently idle at a hub. The operator pool below, by contrast, stays at a constant m servers all day; the two axes are orthogonal by design, the *fleet* sized to demand while the *operator* count that gates it is held fixed for this validation. The incident-hold cap widens during the peaks, when vehicle-kilometres are highest: the incident process acts as a time-varying tax on the active fleet, every held vehicle removed from service until an operator resolves it, while the operator occupancy (bottom) tracks that same demand shape well below its ceiling, so at this staffing the queue rarely binds.

Because incidents are generated per driven metre, they concentrate on the main corridors and around the hubs where vehicle-kilometres are highest, so operator demand is spatially uneven even though the base rate is uniform, reproducing the road network rather than the service-area polygon. Sweeping the number of operators against ADS maturity (which lowers λ) traces how service quality and operator utilisation respond, and confirms the substitution logic: the same served-ride level can be reached either with a mature system and few operators or with a less mature system and many operators.

To ground these dynamics in theory, we run the module in the controlled queueing regime described above (all operators on shift, non-binding κ , and a *single* incident severity class) so that the service-time distribution is exactly the one we prescribe rather than a between-class mixture. Setting that distribution to be exponential collapses the operator pool to an $M/M/m$ system, whose queue delay Erlang-C predicts in closed form. Two rates must be kept apart here: the queue is driven by the *temporal* arrival rate $\Lambda(t)$ of Eq. 2, measured in incidents per unit time, not by the per-distance hazard λ from which it is generated. The offered load in Erlang units is accordingly $a = \Lambda E[S]$, with $E[S]$ the mean incident duration (≈ 84 s here), and $\rho = a/m$ is the load per operator. A single simulated day naturally sweeps that load from near zero at night to its peak, and by scaling the incident rate with the operator count we hold the load per server roughly comparable across the ladder $m \in \{1, 2, 3, 4, 6\}$. Concretely, λ is set so that a vehicle experiences an incident every 91 km at $m = 1$ down to

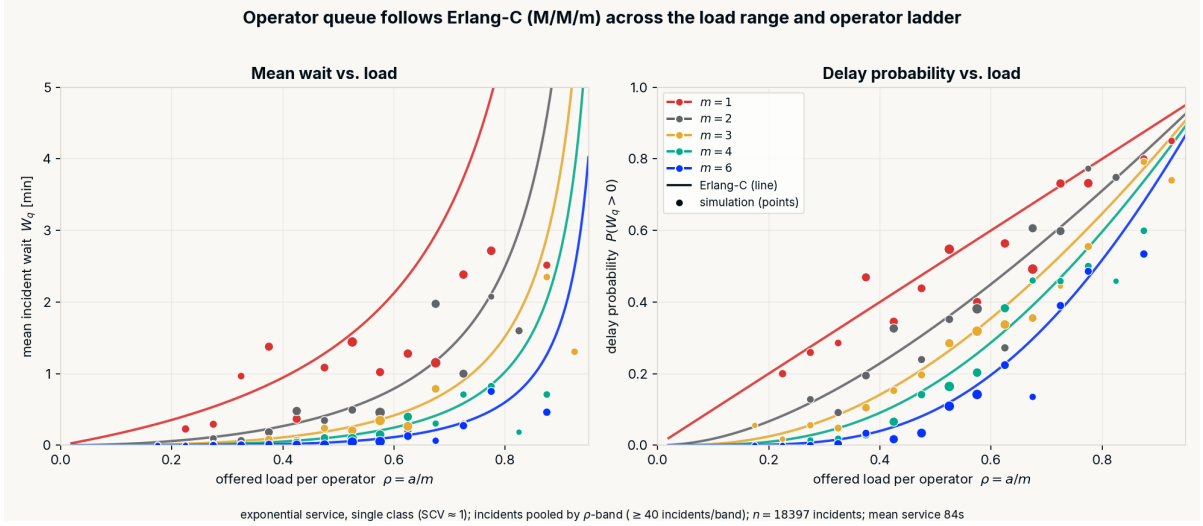


Figure 3: Operator queue against offered load per operator $\rho = a/m$ ($a = \Lambda E[S]$; exponential single-class service; incidents pooled by ρ -band). *Left:* mean wait W_q (axis capped at 5 min). *Right:* delay probability $P(W_q > 0)$. Markers are simulated, lines the closed-form Erlang-C prediction for each $m \in \{1, 2, 3, 4, 6\}$.

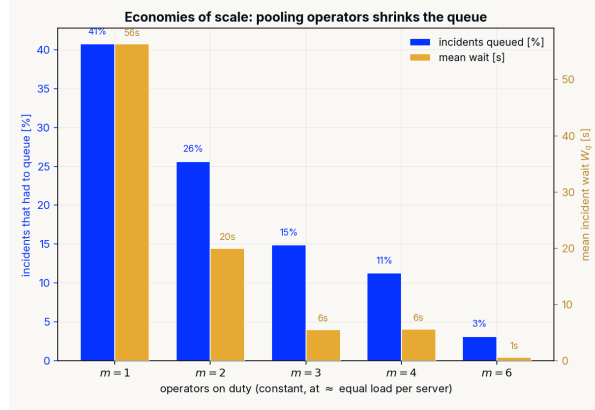


Figure 4: Economies of scale at roughly fixed load per server (one incident every 91 km at $m = 1$ to every 15 km at $m = 6$): raising the operator count from one to six drives the queued share and the mean wait down.

every 15 km at $m = 6$ in the lightest of three load levels, and every 56 km down to every 9 km in the heaviest, a design device to cover the load range at every server count, not a claim about ADS maturity. Tagging each incident with the load ρ prevailing in its arrival bin and pooling incidents into ρ -bands (rather than plotting high-variance per-bin means) yields the saturation curves of Figure 3: for every server count the simulated mean wait W_q (left) and delay probability $P(W_q > 0)$ (right) track the closed-form Erlang-C curve for that m across the whole load range, up to $\rho \approx 0.9$. The family of curves, with higher m lying strictly below, separates the operator counts at equal per-server load, and the upward curvature towards $\rho \rightarrow 1$ is the queueing non-linearity, both reproduced with no free parameters. The one systematic deviation appears only in the mean wait, not in the delay probability. Because the offered load is read off the realised arrivals (the x -axis), any *level* effect, whether from the daily demand profile or from vehicles held off the road throttling their own incident rate, is already absorbed there. What remains as a vertical gap at fixed ρ is that within a day the queue never relaxes to the stationary value for a momentary peak, and that the endogenous arrival process is less bursty than the Poisson stream Erlang-C assumes. Both suppress the long-wait *tail*, so the mean wait falls below Erlang-C (and in the highest ρ -bands even turns down) while $P(W_q > 0)$ stays close to the textbook curve. This softening of the tail is a first instance of the pool outrunning its stationary description, taken up in the conclusions.

The same ladder exposes the economies of scale that make operator pooling attractive. Figure 4 holds the load per server roughly fixed and increases the number of operators from one to six: the share of incidents that must queue falls from over 40% to a few percent, and the mean wait from about a minute to a second. Pooling incidents across more operators reduces contention at equal per-operator load: the economies-of-scale ladder familiar from call-centre staffing, here emergent from the agent-based fleet rather than assumed.

5. Conclusions

We presented a remote-operator shift module for MATSim that makes remote-operator labour a simulation-endogenous quantity. Rather than treating autonomous vehicles as constraint-free, the module represents the realistic middle of the automation transition: a finite pool of operators supervises a dynamically sized set of

vehicles through virtual shifts, a stochastic incident process consumes operator capacity, and guidance requests queue when operators are saturated. By building on and continuing the driver-shift extension of Zwick et al. (2022), the module reuses the established shift, break and hub-return machinery and remains interoperable with the existing DRT and electric-fleet infrastructure, so mixed driver/operator fleets and charging are supported without special cases.

The design deliberately keeps the new behaviour opt-in and event-driven, which makes it straightforward to analyse and to validate. Framing the operator pool as an $M/G/m$ queue links the agent-based model to queueing theory and gives a verification anchor: the simulation reproduces Erlang-C exactly where its assumptions hold. Its value, however, lies in the ways the operator pool *departs* from those assumptions. Erlang-C presumes an arrival stream that is exogenous, stationary and independent of system state; the simulated pool violates each of these, in five ways:

1. **Endogenous, non-stationary arrivals.** The incident rate is not a free parameter but $\Lambda(t) = \lambda \text{ d VKT}/\text{d}t$ (Eq. 2), emerging from demand, routing, empty mileage and time-varying speeds.
2. **Supply-side self-throttling.** Held vehicles stop driving, so the arrival rate falls exactly when the queue lengthens, a stabilising feedback that curbs the Erlang-C blow-up at $\rho \rightarrow 1$ and shows up as the sub-Erlang mean wait in Figure 3. A partial closed-form counterpart does exist for this one effect: the finite-source (machine-repair) queue $M/M/m/N/N$ (Gross et al., 2008), in which each of N sources generates work only while operational, captures exactly this state-dependent arrival rate for a fixed, homogeneous fleet, and benchmarking against it is a natural next step. The simulation goes beyond it by letting the per-source rate itself vary with speed and time and by coupling self-throttling to the other four departures below.
3. **Demand-side pooling efficiency.** At higher demand density, ride-pooling serves more riders per vehicle-kilometre, so vehicle-kilometres, and thus incident load, grow sublinearly in demand; a linear closed-form model over-predicts the operators required at scale.
4. **Heterogeneous, time-varying servers.** Operators follow shift plans with activation and recall dynamics, rather than being a fixed set of identical servers.
5. **Spatial structure.** Incidents concentrate on the corridors and hubs where vehicle-kilometres are highest, which an aggregate single-pool model averages away.

The first two are already borne out by the runs reported here. The third is the economically sharpest: combined with square-root staffing, two stacked sublinearities can make the operators required *per served ride* fall with scale, the central question for a dedicated technical study. Alongside it we will extend the single homogeneous operator role toward the multiple frontline roles observed in practice, refine the uniform hazard into link-specific incident rates so that intervention hotspots become endogenous to network geometry, and add in-field assistance as a second supervision mode in which an operator is physically dispatched to the site. Together these turn the two-point manual-versus-autonomous comparison of Zwick et al. (2022) into a continuous transition path parameterised by the supervision ratio, the incident rate and the operator staffing level.

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